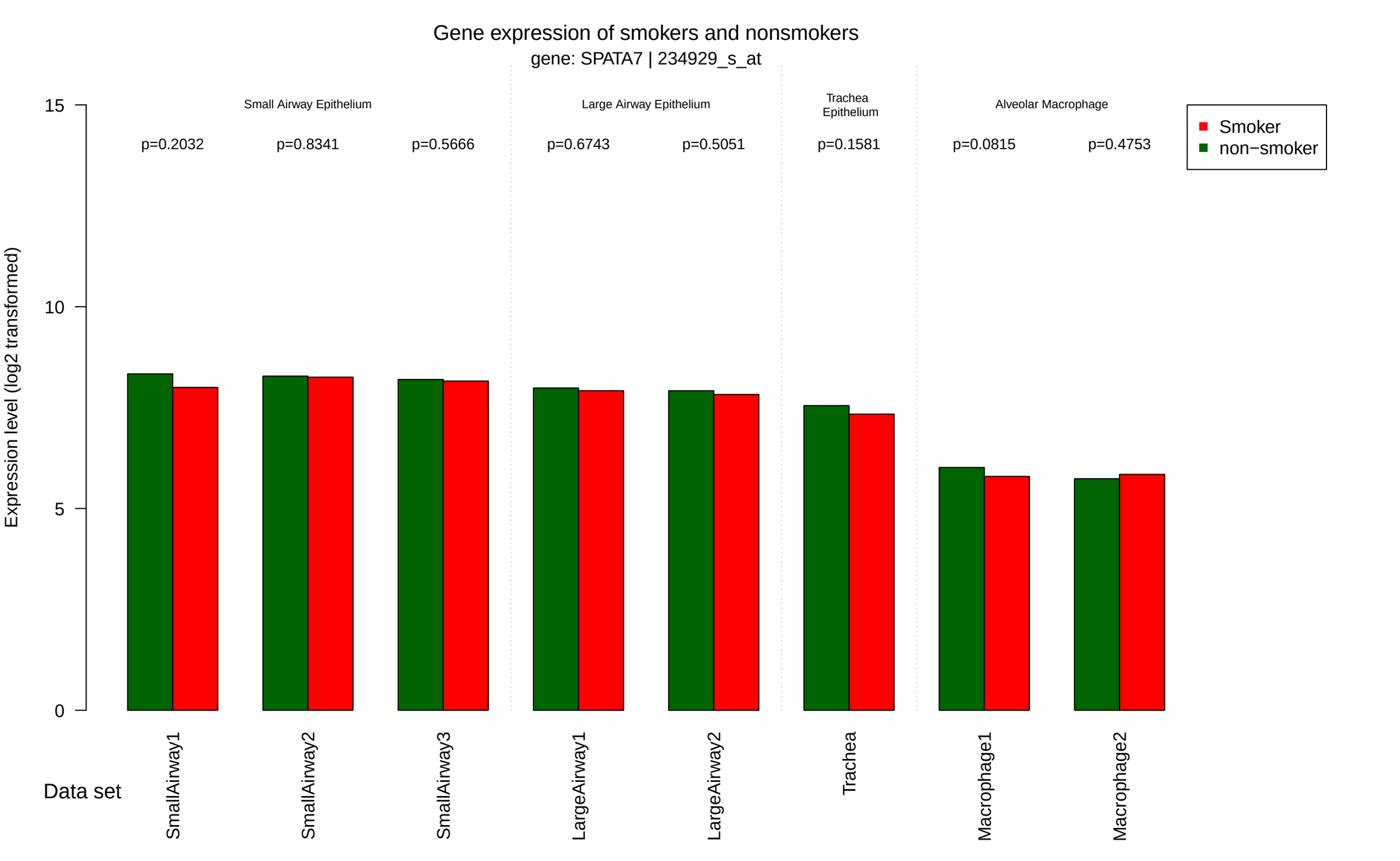
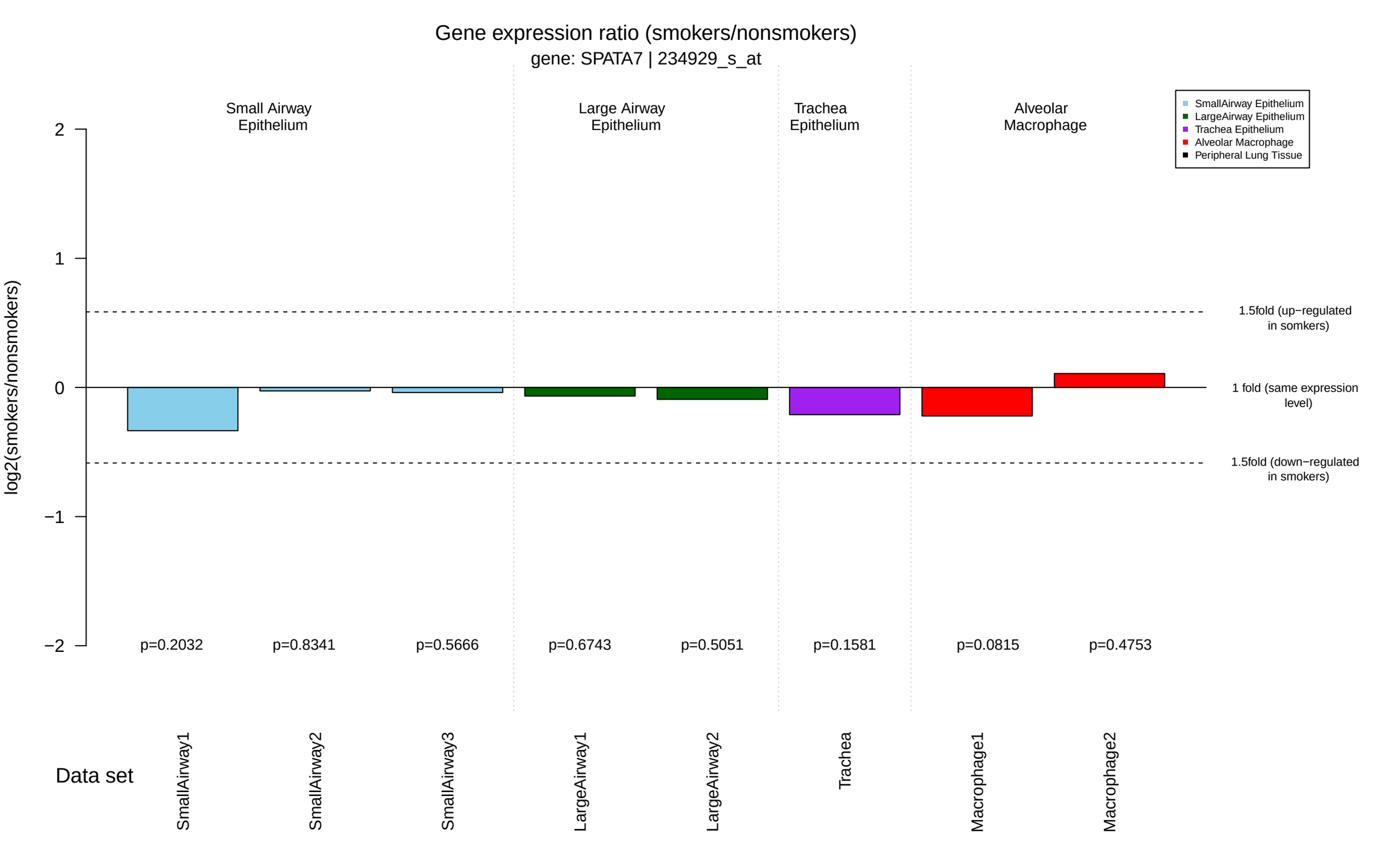
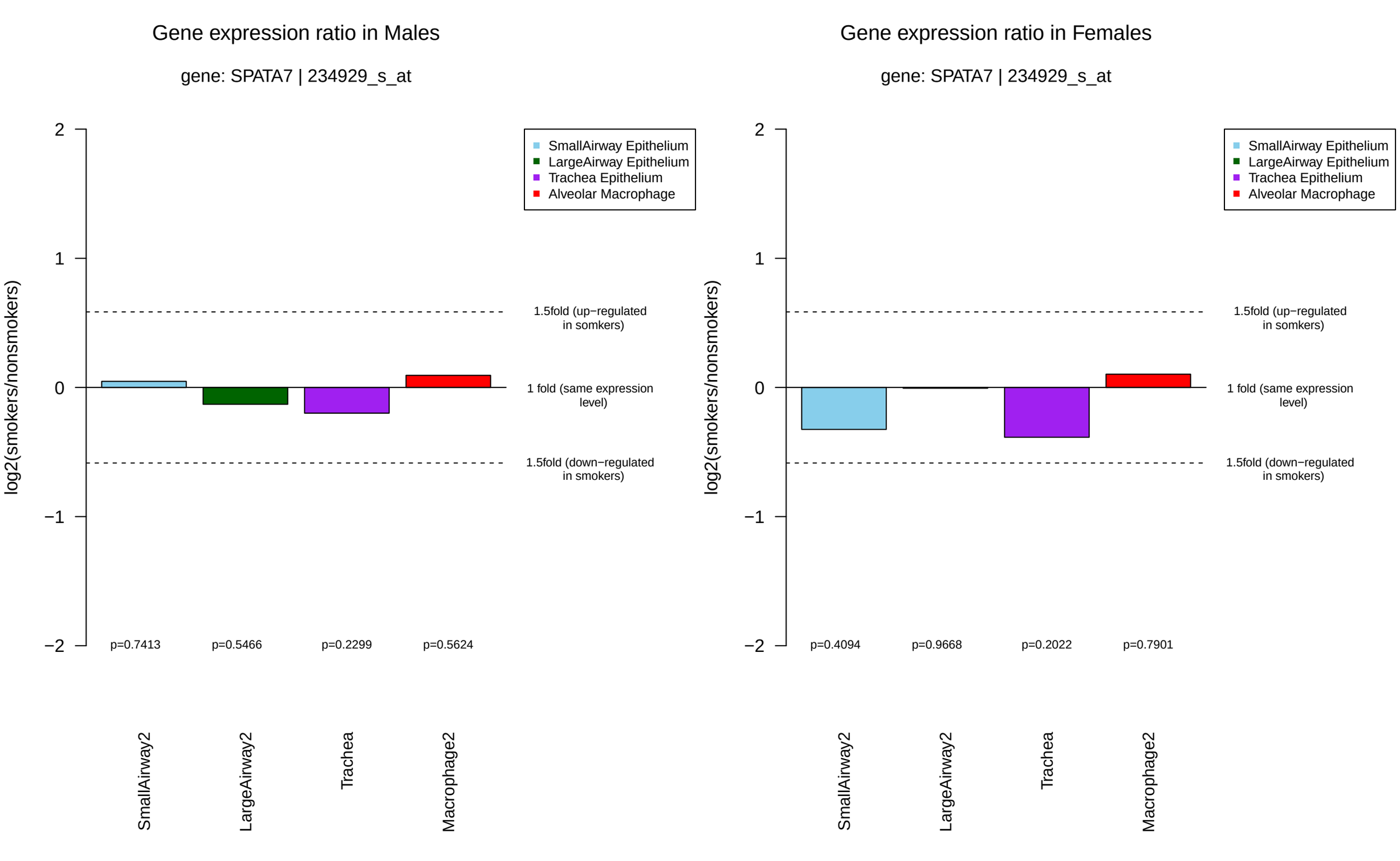
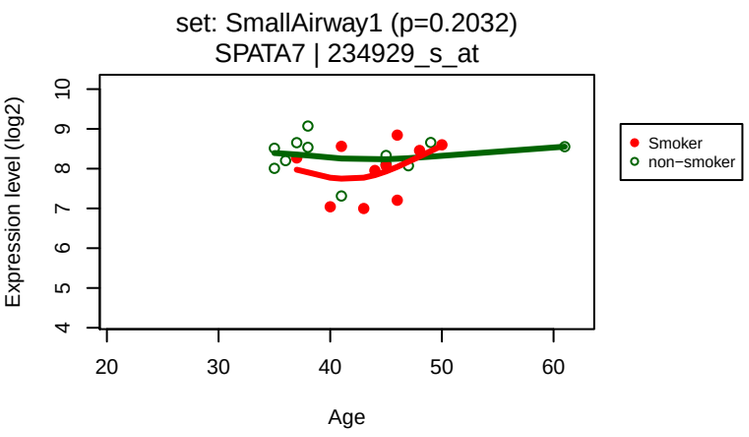
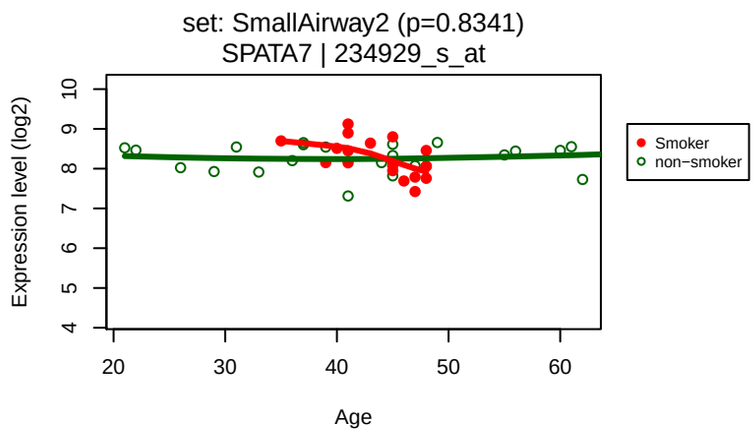
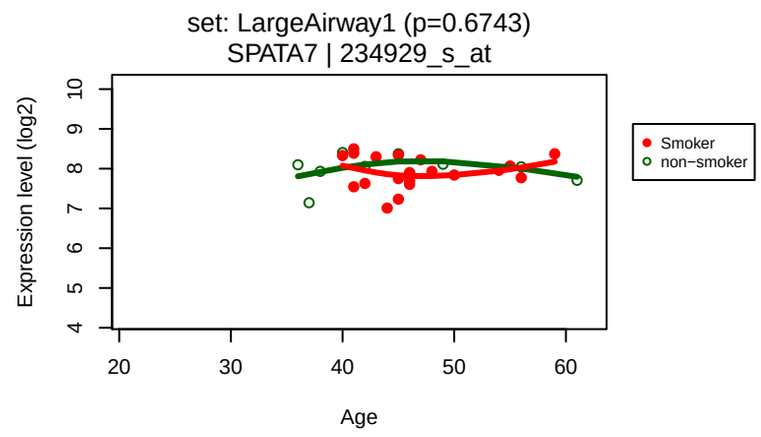

Gene expression of smokers and nonsmokers
gene: SPATA7 | 234929_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2032
p=0.8341
p=0.5666
p=0.6743
p=0.5051
p=0.1581
p=0.0815
p=0.4753
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

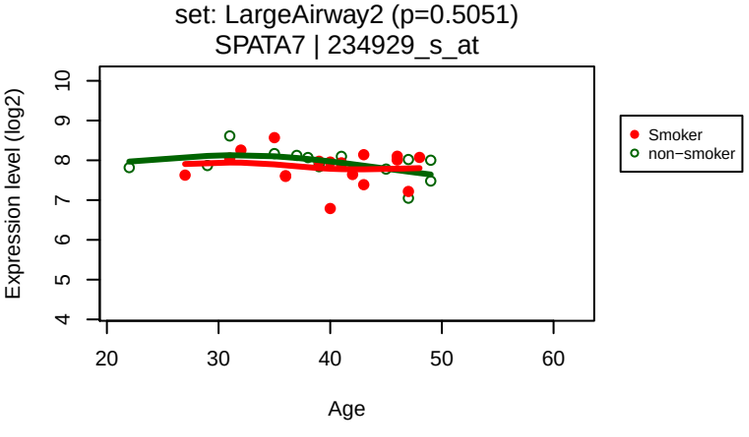

Gene expression ratio (smokers/nonsmokers)
gene: SPATA7 | 234929_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2032
p=0.8341
p=0.5666
p=0.6743
p=0.5051
p=0.1581
p=0.0815
p=0.4753
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

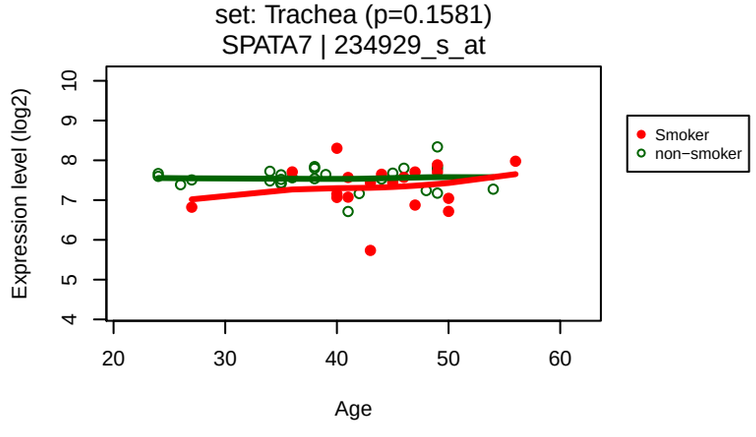

Gene expression ratio in Males
gene: SPATA7 | 234929_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7413
p=0.5466
p=0.2299
p=0.5624
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPATA7 | 234929_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4094
p=0.9668
p=0.2022
p=0.7901
SmallAirway2
LargeAirway2
Trachea
Macrophage2

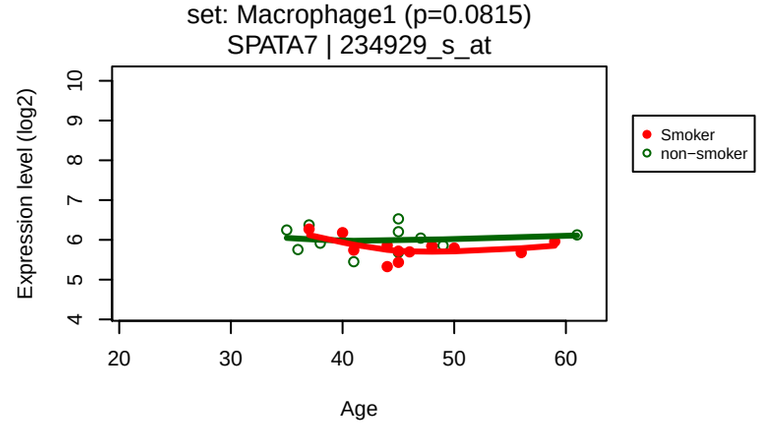

set: SmallAirway1 (p=0.2032)
SPATA7 | 234929_s_at
Expression level (log2)
Age
Smoker
non−smoker

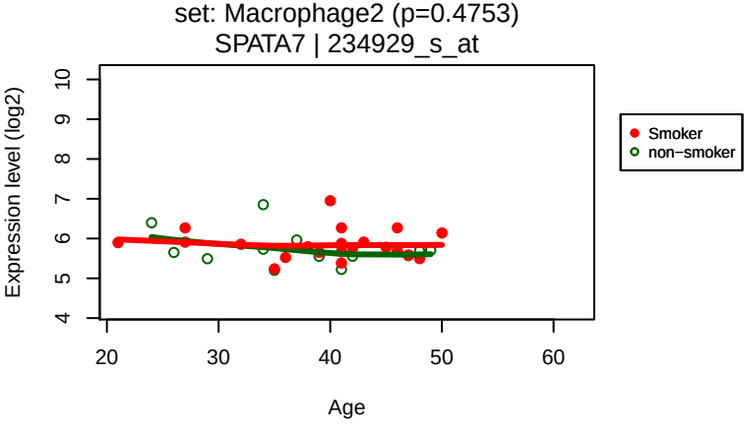

set: SmallAirway2 (p=0.8341)
SPATA7 | 234929_s_at
Expression level (log2)
Age
Smoker
non−smoker

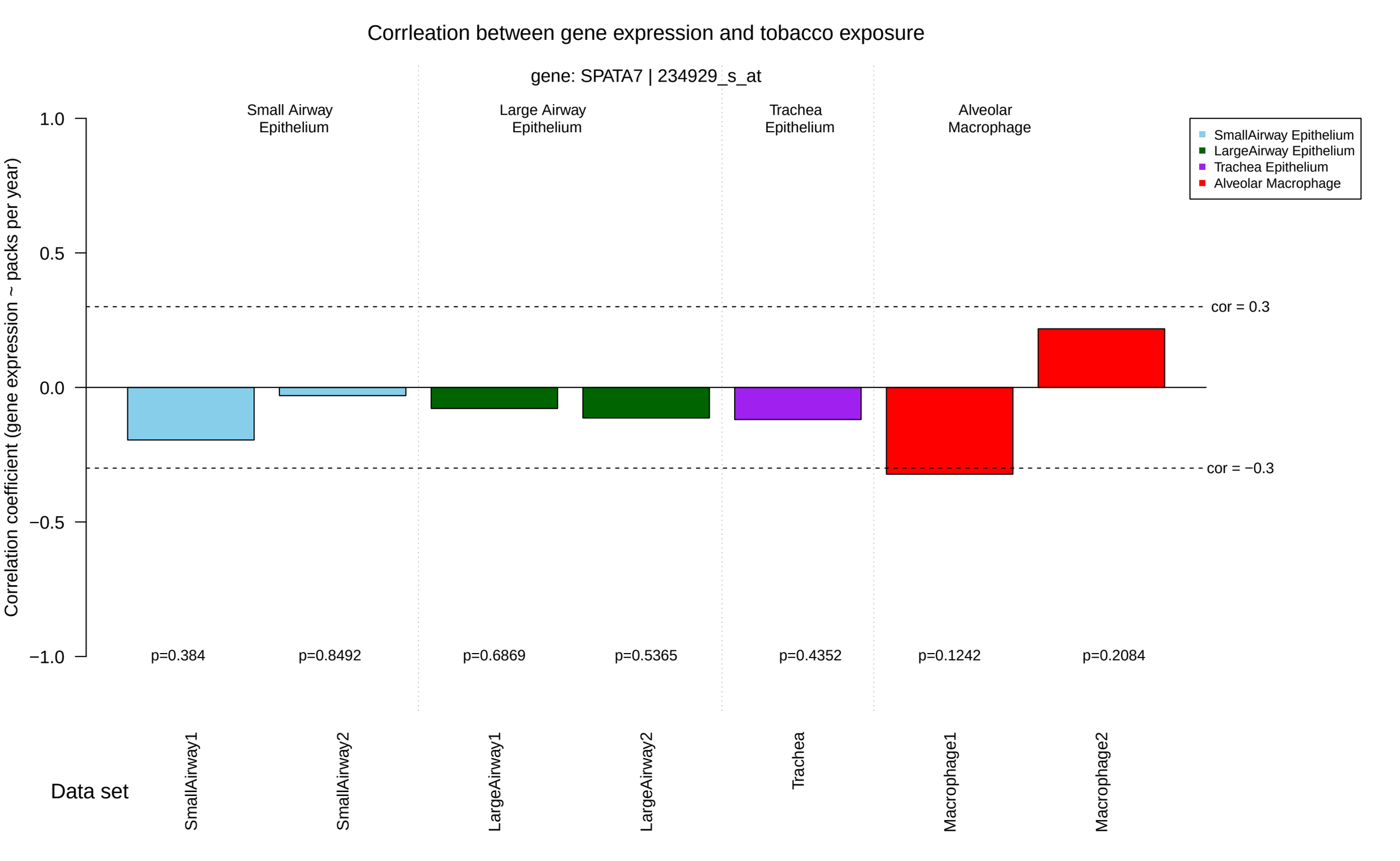

set: LargeAirway1 (p=0.6743)
SPATA7 | 234929_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5051)
SPATA7 | 234929_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1581)
SPATA7 | 234929_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0815)
SPATA7 | 234929_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4753)
SPATA7 | 234929_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPATA7 | 234929_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.384
p=0.8492
p=0.6869
p=0.5365
p=0.4352
p=0.1242
p=0.2084
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2